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TITLE: DevMerge - An Intelligent Collaborative Learning, Practice and Academic Management Platform

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Problem Background

The way we learn in schools and colleges today is often not very effective. There are different tools and systems that are used for learning, testing and evaluating students but they do not work well together. This causes a lot of problems for both students and teachers.

Students have a hard time figuring out what they are bad at, practicing regularly and getting feedback that actually helps them improve. Oftentimes teachers do not have a simple way to give assignments, track progress and understand how well their students are doing.

There are websites and platforms that help with learning and practicing code but they do not provide a complete system that combines learning, testing, working together and analysing results. We need a system that brings all these things together and helps students and teachers in an effective way. The academic ecosystem needs a system that should include learning resources, assessment systems, coding practice platforms and performance evaluation tools.

Objective of the Project

The goal of DevMerge is to create a web platform that brings it all together in an easy and effective manner.

It aims to give institutions an easy-to-use interface for giving tasks, evaluating work and providing feedback. The platform also helps students by giving them an environment to practice coding and see how they are doing in a way that is tailored to them.

The system helps institutions to assign tasks, evaluate work and give feedback in a systematic way. It empowers students with practice environments and performance insights.

DevMerge integrates management, coding practice, collaborative development and AI driven performance analysis.

Proposed Solution

DevMerge is a platform that helps students and educators in ways. It has role-based sections for students and educators and each section has its own set of tools. For educators, DevMerge makes it easy to create assignments, send out notes, manage exams and use artificial intelligence to see how students are doing. For students DevMerge has all sorts of materials and tests to help them learn places to practice coding and a special area where they can work together with others.

One of the stand-out points about DevMerge is that it uses artificial intelligence to look at how users are doing and give detailed reports on what they are good at, what they need to work on and what they should try to improve. DevMerge also has an area called the Playground where students can work together in real time and try out new things, which helps them learn and develop new skills.

Key Features

- Institutional Academic Management System
- Assignment, Test, and Examination Module
- Centralized Notes and Content Distribution
- Integrated Coding Practice Platform (LeetCode-like environment)
- AI-Based Performance Analysis and Feedback System
- Teacher Analytics Dashboard with Student Insights
- DevMerge Playground for Collaborative Coding
- Real-Time Progress Tracking and Performance Monitoring

Technology Used

The system is implemented using a modern full-stack architecture:

- Frontend: React.js, HTML, CSS, JavaScript
- Backend: Node.js with Express.js
- Database: MongoDB
- Architecture: Client–Server Model
- Version Control: Git and GitHub
- Artificial Intelligence techniques are incorporated to analyze user interactions and performance data, enabling dynamic generation of personalized insights.

Expected Outcomes & Benefits

DevMerge will make academic processes more efficient. It does this by bringing everything in one place for learning, practice and evaluation.

Students will get results because they will get continuous feedback and AI will help them learn from their mistakes. Educators will be able to make decisions using the students' performance data. The platform will also help students work together and get better at coding. This will help them not only in their studies but also in their professional growth and careers.

Original Contribution Statement

DevMerge is a way to bring together academic management systems, coding practice, team development tools and smart analytics all in one place.

What makes DevMerge special is that it does not just give students and teachers things to learn and assignments to do. It also gives them feedback that changes based on what they need. This comprehensive approach distinguishes it from existing solutions and

establishes it as a next-generation academic and development platform.

System Architecture

The DevMerge platform follows a client-server architecture where the frontend (React.js) interacts with backend services (Node.js and Express.js), which further communicate with the MongoDB database. An AI module processes user activity data such as coding performance, test scores, and time spent to generate personalized insights.

Module Description

User Authentication Module: Handles secure login and registration.

Assignment & Exam Module: Enables teachers to assign and evaluate tasks.

Coding Practice Module: Provides LeetCode-style problem-solving.

AI Analysis Module: Generates performance insights.

Teacher Dashboard: Displays student analytics and reports.

AI Integration

The AI module analyzes student performance based on metrics such as accuracy, consistency, and time complexity. It identifies weak areas, suggests targeted practice, and provides feedback to improve learning outcomes.

Project Screen Shots



